25901658 **A**

Reg.No : .....

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION APRIL 2026

SECOND SEMESTER

Multi-Disciplinary Courses (MDC) - MG2MDCECT100 - IOT BASED SMART FARMING

(2024 ADMISSION ONWARDS)

Duration: 1 Hour **Maximum Marks: 35**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer any **35** questions

Each question carries **1** marks

1. Which of the following is a key feature of LoRa? [U] / [CO1, CO3]

   a). High power consumption

   b). Short-range communication

   c). Low power consumption and long-range communication

   d). High data transfer speeds

2. Which protocol is commonly used in IoT devices for short-range communication? [K] / [CO1, CO3]

   a). Zigbee

   b). HTTP

   c). SMTP

   d). FTP

3. IoT became commercially significant with the advent of: [K] / [CO1]

   a). Cloud computing

   b). Gaming consoles

c). Virtual assistants d). Smartphones

4. How does the LM35 temperature sensor provide output? [U] / [CO1, CO3]

a). As a digital signal b). As an analog voltage proportional to temperature

c). As a binary value d). Through Wi-Fi transmission

5. What is one advantage of using humidity sensors in greenhouses? [A] / [CO1, CO3]

a). They eliminate the need for ventilation b). They help maintain optimal humidity for plant growth

c). They reduce the need for watering d). They increase the soil pH

6. Which of the following is a common application of Zigbee? [A] / [CO1, CO3]

a). Home automation b). High-definition video streaming

c). Satellite communication d). Online gaming

7. Which of the following sensors is used for detecting rainfall in agricultural IoT applications? [U] / [CO1, CO3]

a). Capacitive Soil Moisture Sensor b). Rain Gauge Sensor

c). Thermocouple Sensor d). Ultrasonic Sensor

8. Microcontrollers enable IoT devices to: [A] / [CO1, CO3]

a). Provide wireless connectivity b). Store multimedia files

c). Display real-time data d). Process data and control actuators

9. What is the primary purpose of IoT devices? [A] / [CO1, CO3]

a). To connect and share data over the internet b). To act as stand alone devices

c). To replace computers d). To provide entertainment only

10. Which of the following facilitates communication between IoT devices? [K] / [CO1]

a). TCP/IP b). SMTP

c). HTTP d). MQTT

11. What is the primary role of sensors in an IoT system? [K] / [CO1, CO3]

a). Collect and transmit data b). Store information

c). Provide user interface d). Control hardware devices

12. How do servo motors function in robotic arms? [K] / [CO1, CO3]

a). By converting electrical signals into data storage b). By rotating to a specified angle based on control signals

c). By providing analog input only d). By detecting weather conditions

13. Microcontrollers communicate with other devices through: [U] / [CO1, CO3]

a). Bluetooth only b). GPIO pins

c). Infrared signals d). Cloud servers

14. Which of the following factors influence the quality of collected agricultural data? [K] / [CO1, CO3]

a). Accuracy of sensors b). Frequency of data collection

c). Proper storage and retrieval of data d). All of the above

15. Smart irrigation systems in farming use IoT to: [K] / [CO1]

a). Automate water supply based on soil moisture levels b). Control livestock movement

c). Detect crop diseases d). Monitor internet usage

16. What is the role of a solenoid valve in irrigation automation? [A] / [CO1, CO3]

a). It collects data from the soil b). It automatically opens/closes to regulate water flow

c). It controls lighting in the greenhouse d). It measures soil moisture levels

17. Which unit is typically used to measure soil moisture content? [A] / [CO1, CO3]

a). Percent (% volumetric water content) b). Parts per million (ppm)

c). Decibels (dB) d). Lumens (lm)

18. Which actuator is used for opening and closing greenhouse windows automatically? [A] / [CO1, CO3]

a). Temperature sensor b). Linear actuator

c). Humidity sensor d). pH control sensor

19. IoT improves efficiency by: [K] / [CO1]

a). Eliminating internet connections b). Reducing data usage

c). Automating routine tasks d). Limiting device connectivity

20. What does IoT mean? [U] / [CO1]

a). Internet of Technology b). Intelligent Online Tracking

c). Internet of Things d). Integration of Tools

21. What is the stage called when a young plant develops its first true leaves? [K] / [CO2]

a). Seedling stage b). Flowering stage

c). Pollination stage d). Fruiting stage

22. How does AI contribute to modern farming? [K] / [CO2]

a). Predicts weather patterns b). Reduces soil nutrients

c). Eliminates crop diseases d). Increases manual labor

23. What are the three primary nutrients required for plant growth? [K] / [CO2]

a). Nitrogen, Phosphorus, Potassium b). Carbon, Oxygen, Hydrogen

c). Calcium, Magnesium, Sulfur d). Iron, Zinc, Copper

24. Why is pH monitoring important in hydroponics? [U] / [CO2]

a). It helps prevent water evaporation b). It ensures proper nutrient absorption by plants

c). It eliminates the need for fertilizers d). It reduces the energy requirements of hydroponic systems

25. What is the advantage of using cloud storage for agricultural data? [U] / [CO2]

a). It provides easy access and scalability for large datasets b). It eliminates the need for farming equipment

c). It stops the need for data analysis d). It slows down the decision-making process

26. What role do drones play in agricultural data collection? [U] / [CO2]

a). They replace farmers entirely b). They act as automated irrigation systems

c). They remove weeds from fields
d). They capture aerial images to assess crop health

27. What is a common challenge in retrieving agricultural data? [U] / [CO2]

a). The fact that modern farmers do not use technology
b). The lack of need for historical data in farming
c). The absence of any data storage systems
d). Inconsistent data storage formats

28. What technology helps optimize irrigation? [K] / [CO2]

a). Excessive pesticide use
b). Manual watering
c). IoT sensors
d). Traditional farming methods

29. What is the process by which a seed begins to grow into a new plant called? [K] / [CO2]

a). Germination
b). Photosynthesis
c). Pollination
d). Transpiration

30. What is the role of artificial intelligence in vertical farming? [U] / [CO2]

a). Monitoring plant growth and adjusting environmental factors
b). Reducing the need for fertilizers
c). Increasing soil moisture
d). Enhancing the taste of crops

31. How does data help in pest management? [U] / [CO2]

a). By predicting pest outbreaks and suggesting timely interventions
b). By completely eliminating pests from farms
c). By increasing pesticide usage without analysis
d). By decreasing the importance of soil health

32. How can Artificial Intelligence (AI) enhance precision farming? [U] / [CO2]

a). By analyzing data patterns and making predictive recommendations
b). By reducing the need for technology in farming
c). By making decision-making more difficult
d). By increasing the use of non-data-driven farming techniques

33. Which pesticide application method minimizes environmental impact? [K] / [CO2]

a). Aerial spraying
b). Over-spraying
c). Targeted application
d). Excessive use

34. Which gas is a major contributor to climate change from farming? [K] / [CO2]

a). Methane b). Oxygen

c). Nitrogen d). Argon

35. Which crop requires the most water to grow? [K] / [CO2]

a). Wheat b). Rice

c). Corn d). Tomatoes

36. What is a common issue faced by traditional farmers? [K] / [CO2]

a). Soil erosion b). Cloud computing

c). AI integration d). Satellite imaging

37. What is the role of genetic modification in plant yield? [K] / [CO2]

a). Increasing disease resistance b). Reducing photosynthesis

c). Eliminating soil nutrients d). Increasing water evaporation

38. Which nutrient is primarily responsible for leaf growth in plants? [U] / [CO2]

a). Nitrogen b). Phosphorus

c). Potassium d). Calcium

39. Which weather factor affects farming the most? [K] / [CO2]

a). Temperature b). Soil pH

c). Pesticide type d). Market prices

40. What is the main difference between aeroponics and hydroponics? [U] / [CO2]

a). Hydroponics uses misted water, while aeroponics uses deep soil irrigation

b). Aeroponics delivers nutrients through misted air, while hydroponics uses a nutrient-rich water solution

c). Hydroponics requires constant sunlight, while aeroponics does not

d). Aeroponics does not require monitoring of oxygen levels, while hydroponics does

[1x35 = 35]

25901658 **B**

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